

ADVANCED LINEAR ALGEBRA

HOMEWORK 5

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P.3.1 Let $X = [X_1 X_2] \in \mathbf{M}_{m \times n}$ in which $X_1 \in \mathbf{M}_{m \times n_1}$ $X_2 \in \mathbf{M}_{m \times n_2}$, and $n_1 + n_2 = n$. Compute $X^\top X$ and $XX^\top$.

Proof. Sol □

P.3.12 Compute

$$\det \begin{bmatrix} 0 & 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 & 3 \\ 3 & 5 & 6 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Solution. Sol □

P.3.26 Let $A, B \in \mathbf{M}_{m \times n}$ and suppose that $r = \text{rank}(A) \geq 1$. Show that the following are equivalent:

- (a) $\text{col}(A) = \text{col}(B)$.
- (b) There are full rank matrices $X \in \mathbf{M}_{m \times r}$ and $Y, Z \in \mathbf{M}_{r \times n}$ such that $A = XY$ and $B = XZ$.
- (c) There is an invertible $S \in \mathbf{M}_n$ such that $B = AS$.

Hint: If $A = XY$ is a full-rank factorization, then $A = [X 0_{n \times (n-r)}]W$, in which $W = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \in \mathbf{M}_n$ is invertible.

Proof. Sol □

P.3.27 Let $A, C \in \mathbf{M}_m$ and $B, D \in \mathbf{M}_n$. Show that there is an invertible $Z \in \mathbf{M}_{m+n}$ such that $A \oplus B = (C \oplus D)Z$ if and only if there are invertible $X \in \mathbf{M}_m$ and $Y \in \mathbf{M}_n$ such that $A = CX$ and $B = DY$.

Proof. Sol □

P.3.29 Let $A, B \in \mathbf{M}_{m \times n}$ and suppose there is a $X \in \mathbf{M}_n$ such that $A = BX$. If $\text{rank}(A) = n$ show that $\text{rank}(B) = n$ and X is invertible. What can you say about the converse?

Proof. Sol

